Ben Boguslavsky

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Education

University of Waterloo Sep. 2022 – May 2027

BASc Candidate for Mechatronics Engineering, ***GPA: 3.95/4.00*** *Waterloo, ON*

- Awards: President's Scholarship of Distinction, President's Research Award
- Certifications: Accelerated Computing with CUDA in C/C++ (NVIDIA), TCPS2, French Immersion

Experience

Humanoid Robotics Software Engineer Co-op May 2026 – Aug 2026

Noble Machines *Sunnyvale, CA*

- Integrated the Acontis EtherCAT Master stack in C++ to replace SOEM, reducing robot initialization time from over 1 minute to 15 seconds and establishing distributed clocks synchronization at 1 kHz
- Brought up XMC4800 firmware (C) with Dave IDE, enabling CAN communication with a custom PCB for a new hand gripper attachment
- Built a Qt/C++ dyno control and data acquisition application with a .NET sidecar, integrating high-frequency torque sensing, magnetic brake & actuator control, live data visualization & recording, and automated chirp and impulse-response testing

Soft Robotics Research Co-op Jan 2026 – May 2026

Hamburg University of Technology *Hamburg, Germany*

- Designed a novel hybrid control framework for a tendon-actuated silicone soft robot integrated with a UR10e arm, leveraging ROS 2 in Python, linear regression models, and P-control, achieving positioning error as low as 2 cm
- Authoring a GAMMAS paper detailing data-driven control strategies for non-linear soft robotic systems

Machine Vision Engineer Co-op May 2025 – Aug 2025

Taymer International Inc. *Markham, ON*

- Accelerated inference of YOLO PyTorch models by 30% using ONNX Runtime with CUDA & TensorRT execution providers, building a C++ pipeline with pre/post-processing to cut deployment time from 5 min to 53 sec and eliminate engine tuning
- Integrated Teensy 4.0 microcontroller with C++ into a $50K AI cable inspection system for real-time PWM lighting control and 24-channel I/O expansion, reading camera-generated PWM signals and interfacing with MFC software via serial communication
- Created a C++/OpenCV cable defect detection module in an MFC application using ONNX-based models, with custom filtering, NMS, and picks-per-inch computation to eliminate the need for manual cable measurements and inspection
- Secured $75K in government funding by leading a research project benchmarking dynamic vs. static input object detection models, analyzing metrics across PyTorch, ONNX, and TensorRT to optimize inference speed and deployment scalability

Robotics Undergraduate Research Assistant May 2024 – May 2025

Active and Interactive Robotics Lab - University of Waterloo *Waterloo, ON*

- Created a real-time collision avoidance system for a KUKA manipulator using Ultralytics YOLO and RealSense depth mapping, streaming spatial telemetry via a custom Java socket pipeline to the robot controller
- Designed and implemented a human-robot interaction experiment to study emotional and physiological responses to collaborative robot behavior, programming trajectory deviations on the Sawyer robot using Python (Pygame) and ROS

Product Development Engineer Co-op Sep 2024 – Dec 2024

Virtek Vision International *Waterloo, ON*

- Built C# scripts and WinForms apps using .NET, REST APIs, and Swagger Codegen with RabbitMQ functionality to demonstrate API integration, enabling clients to create personalized software solutions and reduce support inquiries by 25%
- Migrated mobile app to a Nginx-hosted webserver with a Windows installer using Wix Toolset, establishing multi-device projector access and improving workflow efficiency across large-scale manufacturing environments

Junior Developer in Test Co-op Jan 2024 – Apr 2024

AGF Management Limited *Toronto, ON*

- Built Selenium Webdriver tests with JUnit 5, SQL, and Page Object Model design pattern, doubling overall test coverage
- Optimized test scenarios through SCRUM collaboration with technical experts, reducing testing cycle time by 20%

Projects

Two-Axis Machine Control | *C, STM32, UART, ADC, PlatformIO*

- Programmed a 2-axis STM32 motor control system with ADC-based speed input, limit switch interrupts, and L6470 drivers

Home Security Camera | *C++, ESP32CAM, IR/Ultrasonic Sensors, Servo Motor, 3D Printing, SolidWorks, Fusion360*

- Built an ESP32-CAM system with live streaming, servo panning, IR-based password access, and ultrasonic proximity detection

Virtual Reality Clothes Shopping @ Hack The North | *C#, Meta Quest, Unity, Shopify API*

- Won Best Use of Shopify API & Ubisoft Game Challenge by creating a VR app to try on clothes from online stores

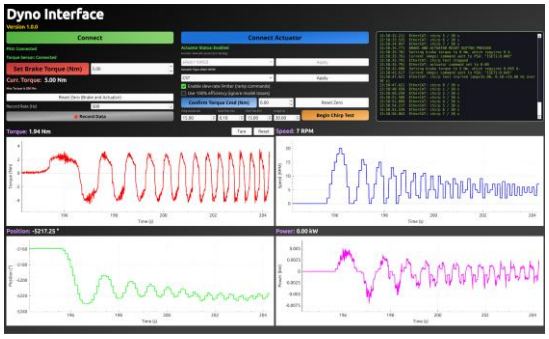
Technical Skills

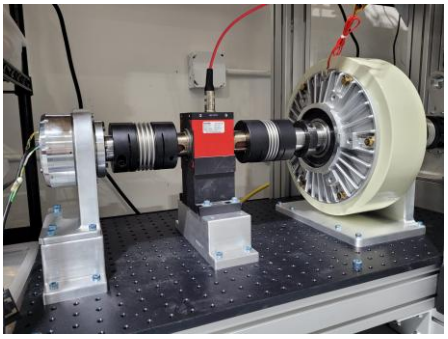
Programming Languages: C++/C, Python, C#, Java, MATLAB, VHDL
Tools & Frameworks: Git, SVN, Linux, ROS/ROS2, OpenCV, CUDA, TensorRT, NumPy, pandas, Selenium WebDriver, JUnit 5, Qt, Microsoft & Atlassian Suite
Embedded & Firmware: STM32, EtherCAT, I2C, SPI, UART, CAN, FPGA

PORTFOLIO BEN BOGUSLAVSKY

Dyno Testbed

- Developed a Qt/C++ application for controlling and testing electric motor systems on a dynamometer, with a .NET sidecar handling supporting high-speed functionality and communication
- Integrated Futek torque sensor hardware and built a 500 Hz data acquisition and recording pipeline for capturing motor and dyno behavior during tests
- Implemented EtherCAT-based motor control and PDO data handling, supporting multiple operating modes including CSP, CSV, and CST, alongside magnetic brake control via external power supply with serial communication
- Built real-time visualization of motor, torque, and control data, with the ability to record test runs for subsequent analysis
- Added user-customizable automated test routines including chirp test up to 100 Hz and impulse-response tests to characterize system dynamics and controller behavior





Two-Axis Machine Control

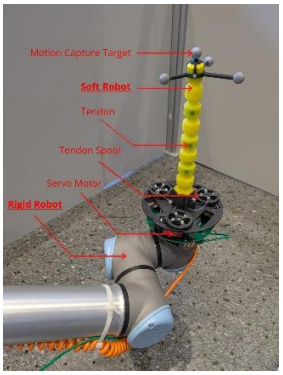
HTTPS://GITHUB.COM/BENBOGUSLAVSKY18/TWO-AXIS-MACHINE-PROJECT/

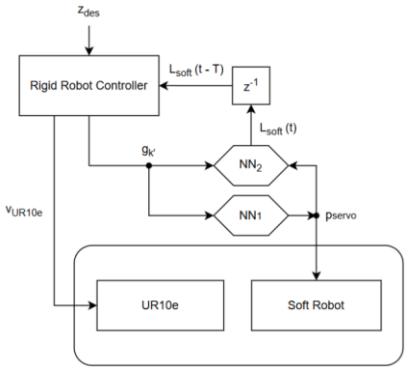
- Programmed real-time control for a two-axis machine using an STM32 microcontroller in C, enabling dynamic speed modulation via analog potentiometers and bidirectional motor control
- Implemented interrupt-driven limit switch handling, modular ADC channel reading, and UART-based debugging for robust system feedback and safety
- Applied embedded systems and microprocessors principles from coursework including ISRs, timers, GPIO (open-drain/push-pull), and structured interfacing with external motor drivers (L6470)

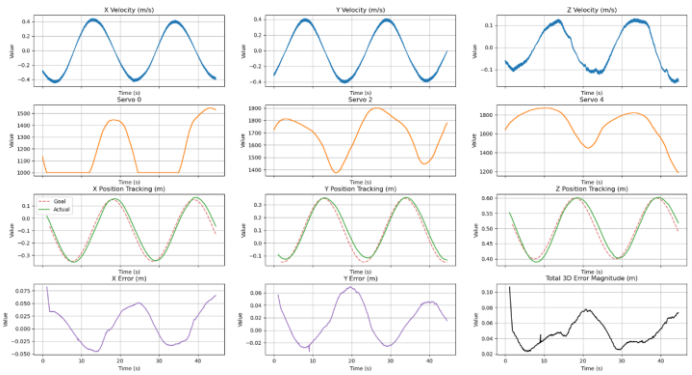


Hybrid Rigid-Soft Robotics Trajectory Tracking Research

- Designed a novel, data-driven hybrid control framework for a tendon-actuated silicone soft robot integrated with a UR10e arm, leveraging ROS 2 in Python, linear regression models, and P-control, achieving positioning error as low as 2 cm



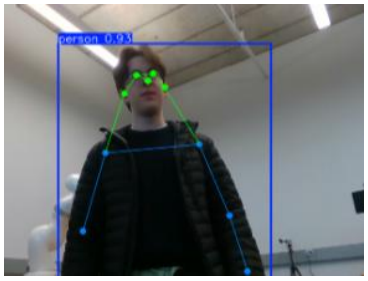
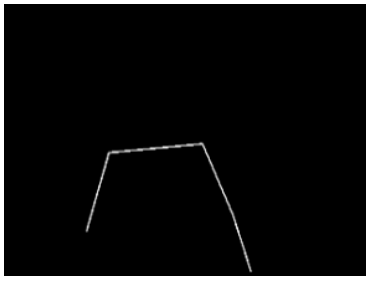




Undergraduate Robotics Research Project 2

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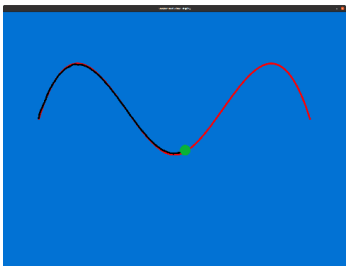
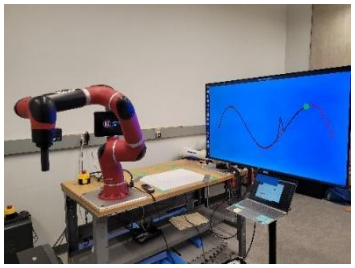
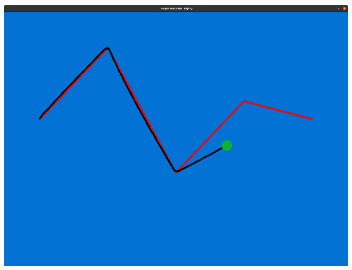
- Created a collision avoidance system for a KUKA robotic arm using 2D pose detection from Ultralytics YOLO (python) processed in OpenCV and RealSense depth camera to track human position, sending spatial info in Java to the KUKA controller over UTP
- Designed a 3D-printable end-effector attachment in SolidWorks to connect with a human model leg, enabling ankle support and knee repositioning for ACL surgery simulation


Undergraduate Robotics Research Project 1

HTTPS://GITHUB.COM/BENBOGUSLAVSKY18/SAWYER_RESEARCH_IMPLEMENTATION

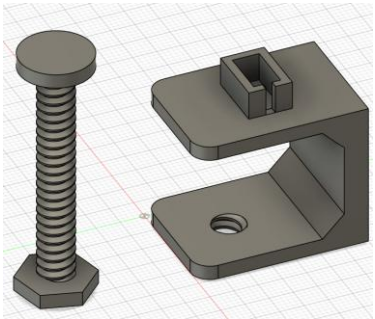
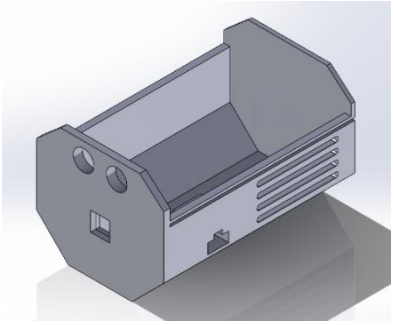
- Designed and implemented a human-robot interaction experiment to study participant responses to autonomous versus manual arm control and motion accuracy, focusing on perceptions of trust, safety, and intent in robotic behavior
- Leveraged ROS and Python on Sawyer robot arm to coordinate spline arm movements along randomized and pre-defined paths, with intentional error in motion to study participant's reaction.
- Captured GSR data with Shimmer3 GSR+, analyzed results using pandas
- Developed a real-time visualization for trial participants using Pygame, dynamically displaying the arm's motion and path for enhanced experimental monitoring and participant interaction

Home Security Camera

HTTPS://GITHUB.COM/BENBOGUSLAVSKY18/SECURITY-CAMERA-PROJECT

- Implemented ESP32Cam (C++) locally hosted web server for real-time video
- Incorporated and programmed sensing components such as ultrasonic sensor, servo motor and IR remote sensor for functionality including controlled camera panning, password authentication, and person detection
- Designed and modelled lightweight, 3D-printable camera shell and desk clamp using SolidWorks and Fusion360


Hack The North 2024 – VR Clothes Shopping

HTTPS://DEVPOST.COM/SOFTWARE/SHOP-THE-NORTH

- Designed and developed a VR application simulating online clothes shopping, utilizing C# and Unity
- Integrated Shopify API to create a dynamic and interactive shopping environment, enabling users to see online store items and try them on
- Shopify API Challenge (Best use of API) and Ubisoft Game Dev Challenge Winner





Toyota Innovation Challenge – Hole/Sticker Detector

HTTPS://GITHUB.COM/BENBOGUSLAVSKY18/TOYOTACHALLENGE-STICKERS

- Worked in teams to develop an AI program to detect/differentiate between holes and stickers on extrusions with 98% accuracy
- Employed Jupyter Notebooks as a primary tool for developing and training a CNN, utilizing Python, Keras and OpenCV
- Engaged in peer-to-peer learning and knowledge sharing, actively seeking feedback from teammates and incorporating suggestions in brainstorming sessions

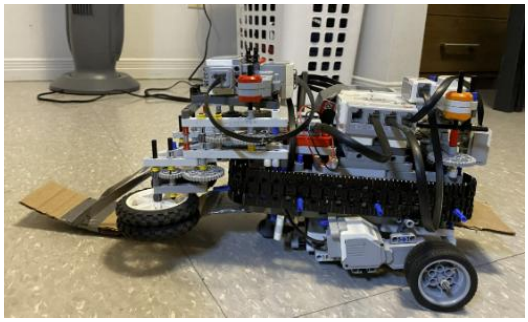




Tennis Training Robot

HTTPS://GITHUB.COM/BENBOGUSLAVSKY18/TENNISTRAININGROBOTPROJECT

- Programmed and built prototype tennis training robot that positions itself in various positions on a tennis court and launches tennis balls in random directions
- Constructed mechanical components such as geared flywheels, intake system and internal conveyors using Lego EV3 Robotics Kit
- Implemented automated runtime functionality and initial mode selection user interface using RobotC
- Developed a color detection mechanism, enhancing system safety and preventing insertion of unauthorized hazardous objects
- Led and collaborated within an Agile environment, participating in sprint planning/reviews while completing deliverables on time

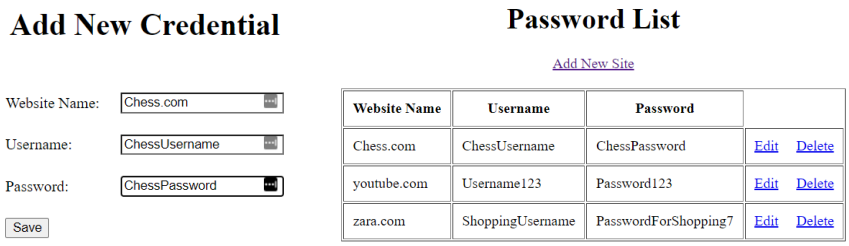




Website Credential Storage

HTTPS://GITHUB.COM/BENBOGUSLAVSKY18/PASSWORDCARDS

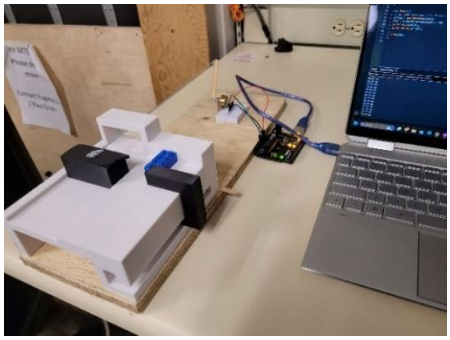
- A Maven-based locally hosted website developed with Java to store username and password credentials, implementing REST APIs and Spring Boot 2
- Applied OOP principles from university course topics

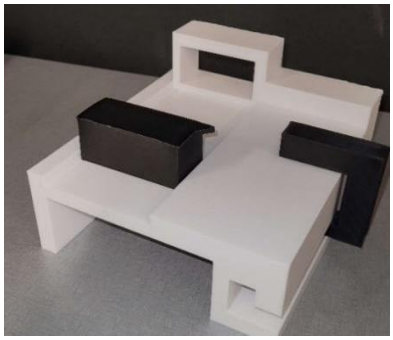


Digital Measurement Device

HTTPS://GITHUB.COM/BENBOGUSLAVSKY18/MEASUREMENT-DEVICE

- Designed, modelled and 3D printed parts for a precision measurement device created using statistical methods using an Arduino with a rotary potentiometer, successfully achieving measurement accuracy within one millimeter
- Conducted data acquisition, calibration and uncertainty analysis to maximize measurement precision and consistency





Electric Motorcycle Design Team – UW Electrium Design Team

- Participated in the design, CAD (SolidWorks) and machining of Electric Motorcycle
- Researched, documented and conducted analysis of several important processes, such as FEA and fairing manufacturing for improvement of future design iterations
- Conducted FMEA, identifying potential failure modes to maximize safety and robustness of motorcycle components

